

NUTRITION ASSISTANT

A Full-Stack Nutrition Tracking Web Application (MERN Stack)

Final Project Report

Submitted as part of project completion requirements

Table of Contents

- 1. Introduction
 - 1.1 Project Overview
 - 1.2 Purpose
- 2. Ideation Phase
 - 2.1 Problem Statement
 - 2.2 Empathy Map Canvas
 - 2.3 Brainstorming
- 3. Requirement Analysis
 - 3.1 Customer Journey Map
 - 3.2 Solution Requirement
 - 3.3 Data Flow Diagram
 - 3.4 Technology Stack
- 4. Project Design
 - 4.1 Problem Solution Fit
 - 4.2 Proposed Solution
 - 4.3 Solution Architecture
- 5. Project Planning & Scheduling
 - 5.1 Project Planning
- 6. Functional and Performance Testing
 - 6.1 Performance Testing
- 7. Results
 - 7.1 Output Screenshots
- 8. Advantages & Disadvantages
- 9. Conclusion
- 10. Future Scope
- 11. Appendix

1. Introduction

1.1 Project Overview

Nutrition Assistant is a full-stack, MERN-based (MongoDB, Express.js, React.js, Node.js) web application designed for health-conscious individuals, students, working professionals, and fitness enthusiasts who want a simple way to track calories and understand what they eat. The platform provides a searchable food and nutrition database, daily calorie tracking, and macronutrient analysis (protein, carbohydrates, fat), enabling users to make informed, data-driven decisions about their diet.

Through a simple registration and secure login process, users gain access to a personalized dashboard where they can search food items, log their daily intake, and view a real-time summary of calories consumed and macronutrient distribution. The application is designed to be lightweight and approachable, so that both nutrition beginners and more experienced trackers can build consistent, healthy habits.

Nutrition Assistant combines a clean, intuitive interface with reliable nutrition data and simple tracking tools to help users build sustainable eating habits over time.

1.2 Purpose

The purpose of Nutrition Assistant is to remove the friction from everyday nutrition tracking by giving users a single, reliable place to search food, log meals, and see the impact on their daily goals. Specifically, the platform aims to:

- Provide a fast, searchable database of food items with accurate nutritional values.
- Let users log daily food intake and see calorie totals update in real time.
- Break down macronutrient consumption (protein, carbohydrates, fat) to support balanced eating.
- Present a clear, at-a-glance dashboard summarizing daily nutrition progress.
- Reduce reliance on manual spreadsheets, paper logs, or scattered nutrition websites.
- Build user confidence and consistency in maintaining healthy eating habits over the long term.

2. Ideation Phase

2.1 Problem Statement

“Health-conscious individuals, students, and working professionals struggle to track their daily calorie intake and understand the nutritional value of the foods they eat — most rely on manual calculations, paper notebooks, spreadsheets, or complex, cluttered apps that are time-consuming and hard to maintain. This lack of a simple, accessible tracking solution makes it difficult for people to build and sustain healthy eating habits.”

This problem affects a wide audience: students trying to eat better on a budget, professionals with little time to track meals manually, fitness enthusiasts monitoring macros, and anyone managing weight loss, weight gain, or a general balanced diet.

2.2 Empathy Map Canvas

An empathy map was created around the persona 'Priya', a working professional who wants to eat healthier but struggles to find time for manual nutrition tracking, to better understand user needs, pain points, and motivations.

SAYS	"I want to eat healthier, but I never know what's actually in my food." "I wish tracking calories didn't feel like a chore." "I need something quick I can check between meetings."
THINKS	"Am I getting enough protein?" "Is this app going to be as complicated as the last one I tried?" "I just want simple, honest numbers."
DOES	Googles calorie counts, screenshots nutrition labels, tracks meals inconsistently in notes apps, follows fitness influencers for tips.
FEELS	Motivated to improve but easily discouraged by complicated tools; relieved when something is fast and clear; proud when a daily goal is hit.
PAINS	Manual calorie math, inconsistent data across apps, no time for detailed logging, information overload from conflicting nutrition advice.
GAINS	A single reliable place to search food and log meals, clear daily/weekly progress, confidence in food choices, better long-term habits.

2.3 Brainstorming

Based on the problem statement and empathy map, the team brainstormed the following ideas, which were then prioritized and grouped into the final feature set of Nutrition Assistant:

- Provide a searchable food database so users can quickly find nutritional information.
- Let users log daily food intake with quantity/serving size to calculate accurate totals.
- Automatically calculate and display calorie and macronutrient totals after each log.
- Build an interactive dashboard summarizing daily nutrition progress at a glance.
- Secure user accounts with registration, login, and protected routes.
- Keep the UI simple and intuitive so tracking feels quick, not like a chore.
- Store all data reliably in the cloud (MongoDB Atlas) so progress is never lost.

These ideas were consolidated into the core modules of Nutrition Assistant: User Authentication, Food Search, Calorie Tracking, Macronutrient Analysis, and the Dashboard.

3. Requirement Analysis

3.1 Customer Journey Map

The customer journey map traces a typical user's experience from first discovering Nutrition Assistant to becoming a consistent, daily user:

Stage	User Actions & Experience
Awareness	User discovers Nutrition Assistant while looking for a simple way to track daily food intake.
Registration	User signs up with name, email, and password; receives confirmation and is redirected to Login.
Onboarding	User logs in and lands on a personalized dashboard showing today's nutrition summary.

Exploration	User searches the food database and reviews nutritional details of items they eat regularly.
Engagement (Logging)	User logs a food item with quantity; the entry is saved and totals update immediately.
Tracking & Analysis	User reviews daily calorie totals and macronutrient breakdown on the dashboard.
Retention	User returns daily to log meals, monitor progress, and build consistent healthy habits.

3.2 Solution Requirement

Functional Requirements:

- User registration and secure login/authentication.
- Personalized user dashboard displaying calorie totals, macronutrient breakdown, and progress.
- Search and browse a comprehensive food database with nutritional details.
- Log daily food intake with quantity/serving size.
- Automatically calculate calories and macronutrients (protein, carbs, fat) consumed.
- Persist user and nutrition data securely using MongoDB Atlas.

Non-Functional Requirements:

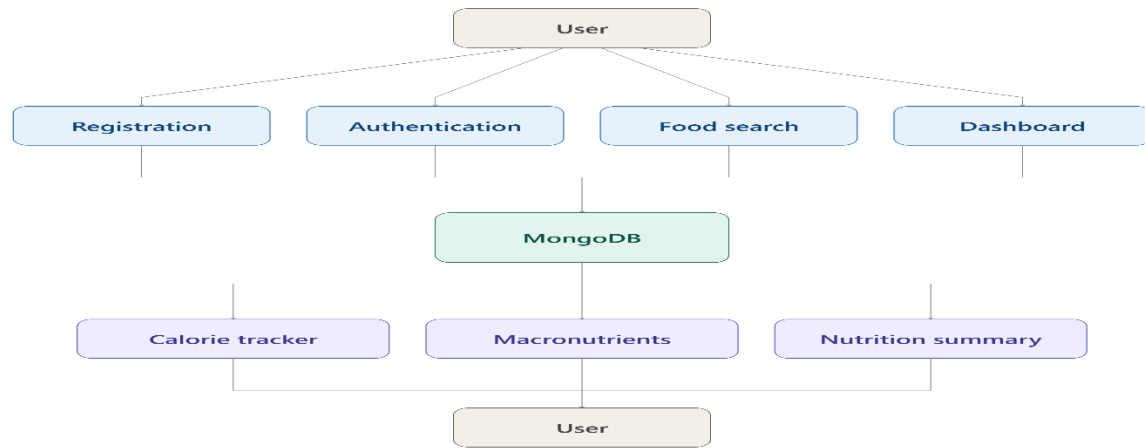
- Usability: Simple, intuitive UI usable by both nutrition beginners and experienced trackers.
- Performance: Fast page loads and near-instant updates to calorie/macro totals after logging food.
- Security: Encrypted password storage (bcrypt) and JWT-protected API routes.
- Scalability: Modular MVC architecture that supports adding new features without major rework.
- Reliability: Consistent, accurate recalculation of nutrition totals after every log entry.
- Availability: The platform should remain accessible with minimal downtime.

3.3 Data Flow Diagram

Level 0 (Context Diagram):



Level 1 (Detailed DFD): The system is broken down into core processes that map directly to the backend controllers:



3.4 Technology Stack

Layer	Technology	Purpose
Frontend	React.js	Building dynamic, responsive UI components (dashboard, food search, charts).
Frontend Tooling	Vite, Tailwind CSS, npm	Fast project scaffolding, utility-first styling, and dependency management.
Backend	Node.js, Express.js	Server-side runtime and RESTful API routing/business logic.
Database	MongoDB Atlas (Mongoose)	Cloud NoSQL storage for users, food items, and nutrition records.
Authentication	JWT + bcrypt	Securing protected routes and hashing user passwords.
API Testing	Postman	Testing and validating backend REST APIs during development.
IDE	Visual Studio Code	Primary code editor with integrated Git and terminal support.
Deployment	Vercel (frontend), Render (backend)	Hosting the production frontend and backend services.
Version Control	Git & GitHub	Source code management and collaboration.

4. Project Design

4.1 Problem Solution Fit

Problem: Health-conscious individuals lack a simple, accessible way to track calorie intake and understand the nutritional value of their food.

Solution Fit: Nutrition Assistant directly addresses this by combining a searchable food/nutrition database with instant calorie and macronutrient calculation, so users get accurate feedback the moment they log a meal. The dashboard closes the feedback loop by summarizing daily progress,

which is exactly the quick, low-friction tracking cycle identified as the core user need in the empathy map.

4.2 Proposed Solution

Problem Statement	Health-conscious individuals lack a simple, accessible way to track calories and understand the nutritional value of what they eat.
Idea / Solution	A MERN-based web application (Nutrition Assistant) offering food search, calorie tracking, macronutrient analysis, and an interactive nutrition dashboard.
Novelty / Uniqueness	Combines a searchable food/nutrition database with real-time calorie and macronutrient calculations in a single lightweight dashboard, rather than a spreadsheet or scattered apps.
Social Impact / Value	Improves nutrition awareness and supports healthier eating habits, helping users build sustainable, informed routines.
Business / Scalability Model	Can be extended with premium meal planning, AI-based recommendations, or partnerships with nutrition/fitness brands.
Scalability of the Solution	Modular MVC backend and component-based React frontend allow new features (barcode scanning, meal planner, reports) to be added with minimal rework.

4.3 Solution Architecture

Nutrition Assistant follows a modular full-stack architecture using the MERN stack, cleanly separating the Frontend (React), Backend (Express + Node.js), and Database (MongoDB Atlas) layers:

- Frontend: Built using React, Tailwind CSS, and Vite, providing Registration, Login, Dashboard, Food Search, Calorie Tracker, and Macronutrient Dashboard pages.
- Backend: Built using Node.js and Express, exposing REST API endpoints for Authentication, Users, Food Search, and Dashboard data, secured with JWT authentication.
- Database: MongoDB Atlas stores collections for Users, Food, and Nutrition Records.

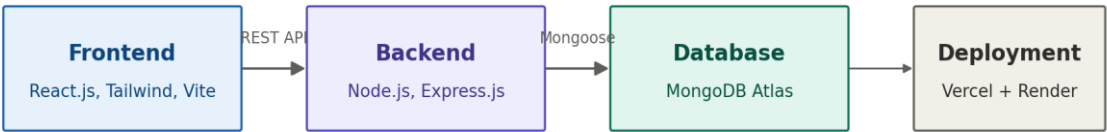


Fig 4.1 – Nutrition Assistant Technical Architecture Diagram

The corresponding ER diagram below shows how the User, Food, and NutritionRecord entities relate: a User can create many NutritionRecord entries; each NutritionRecord links a User to a Food item along with the logged quantity and date.

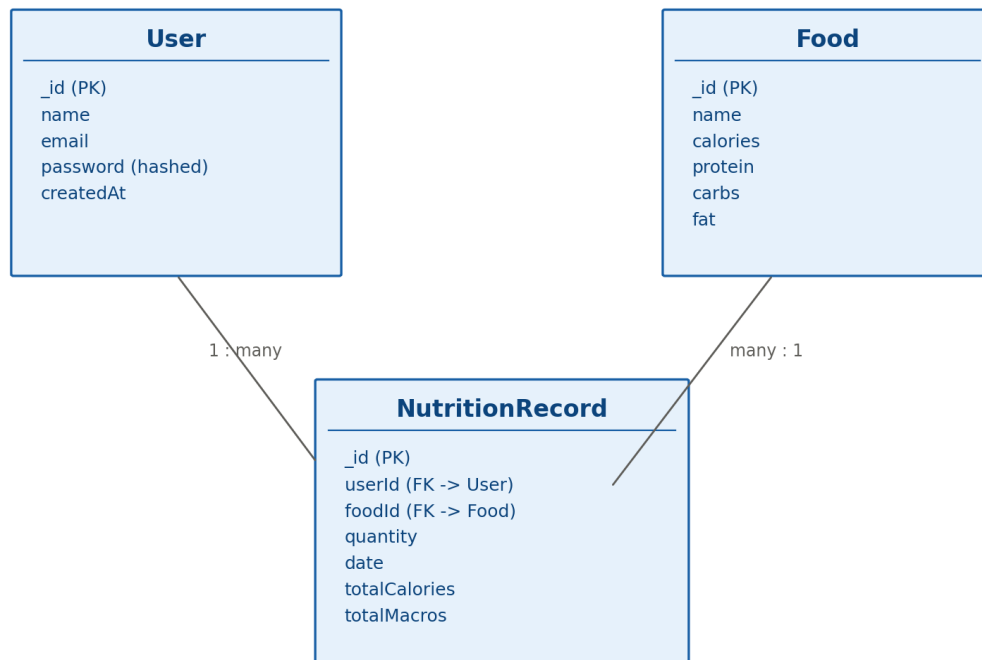


Fig 4.2 – Nutrition Assistant ER Diagram

The diagram below traces the typical end-to-end user flow through the application, from first visiting the site to logging out.

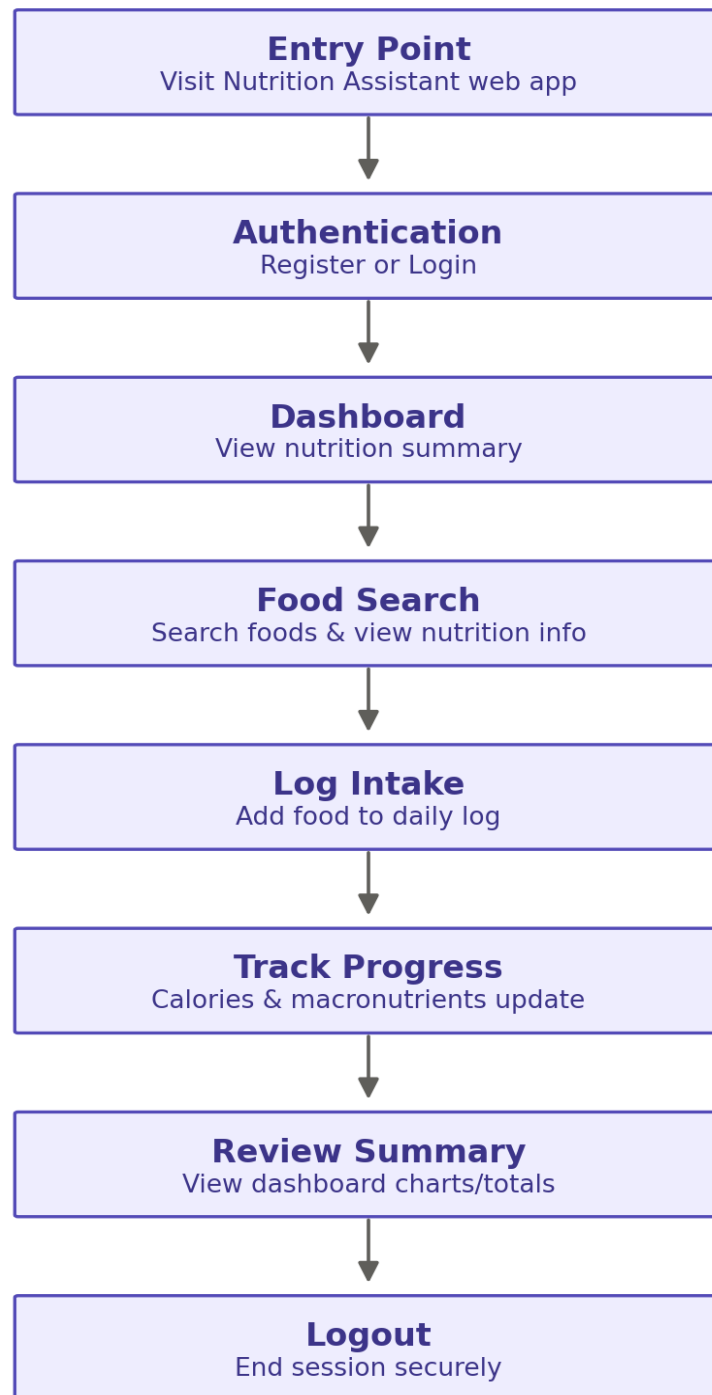


Fig 4.3 – Nutrition Assistant User Flow

5. Project Planning & Scheduling

5.1 Project Planning

The project was planned and executed in an iterative, sprint-based manner, moving from setup and core backend/database design through frontend development, integration, and final testing:

Sprint	Functional Requirement (Epic)	Tasks	Story Points
Sprint 1	Project Setup & Database Design	Set up client/server folders; initialize React (Vite) and Express projects; configure MongoDB Atlas connection; design schemas (User, Food, NutritionRecord).	10
Sprint 2	User Authentication	Implement registration & login APIs; build JWT authentication middleware; create Register/Login UI components.	10
Sprint 3	Food Search & Nutrition Data	Build Food model/controller/routes; integrate nutrition database lookups; build Food Search UI and results view.	10
Sprint 4	Calorie Tracking & Macronutrient Dashboard	Implement daily logging of food intake; calculate calorie and macronutrient totals; build Dashboard UI with summaries.	10
Sprint 5	Testing & Deployment	Conduct functional and performance testing; fix bugs; deploy frontend to Vercel and backend to Render.	10

Each sprint followed a consistent cycle of planning, development, internal review, and testing before moving to the next stage, ensuring incremental delivery of working features.

6. Functional and Performance Testing

Functional Testing: Each module — user registration/login, food search, food logging, calorie calculation, macronutrient breakdown, and dashboard reporting — was tested against its expected behavior using both manual testing in the browser and API testing in Postman. Test cases covered valid/invalid login credentials, successful and failed food logging (e.g., invalid quantity), correct recalculation of totals after each log, and correct authorization checks on protected routes. A detailed User Acceptance Testing report with individual test cases and results is maintained separately.

6.1 Performance Testing

Performance testing focused on response times and data consistency under typical usage:

API Response Time	Core APIs (login, food search, logging intake) responded within acceptable limits during manual and Postman-based checks.
Dashboard Update Speed	Calorie and macronutrient totals on the dashboard updated promptly after each logged food item.
Concurrent User Handling	Multiple simulated user sessions (registration, login, food logging) were tested to ensure MongoDB and Express handled concurrent requests without data corruption.
Frontend Load Time	The React app (built with Vite) loaded key pages such as Dashboard and Food Search quickly on standard hardware/browsers.

Database Query Performance	Mongoose queries for fetching food items and nutrition records were optimized to return results efficiently for typical dataset sizes used in testing.
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7. Results

7.1 Output Screenshots

The following screenshots demonstrate the working Nutrition Assistant application across its key screens. Placeholder images are shown below

Fig 7.1 – Landing / Home Page

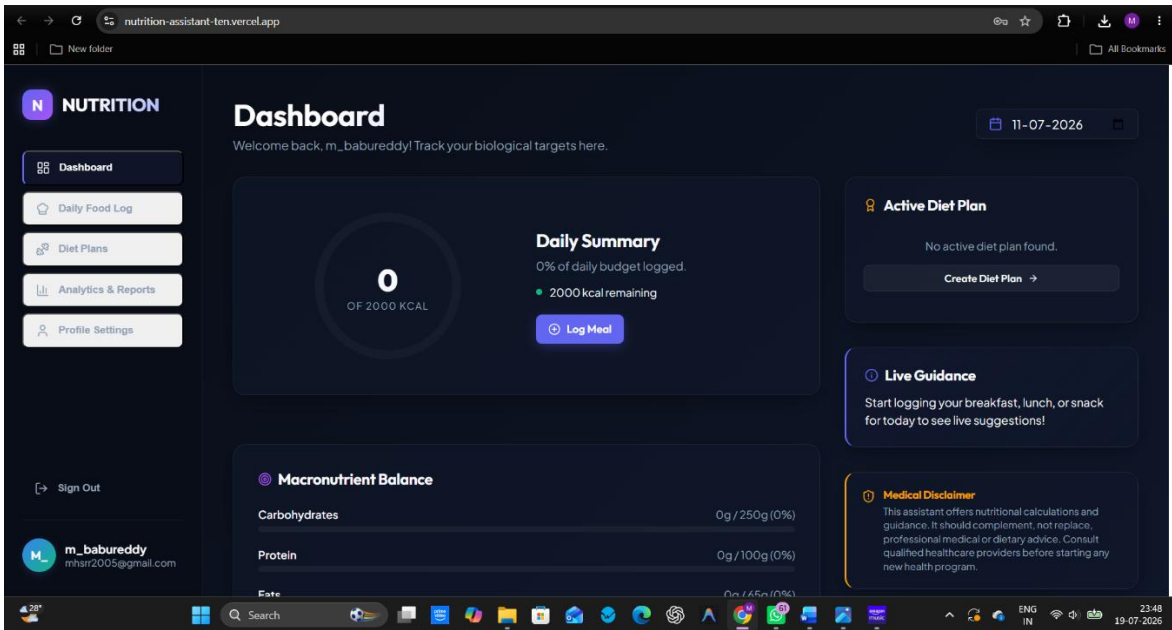


Fig 7.2 – Registration Page

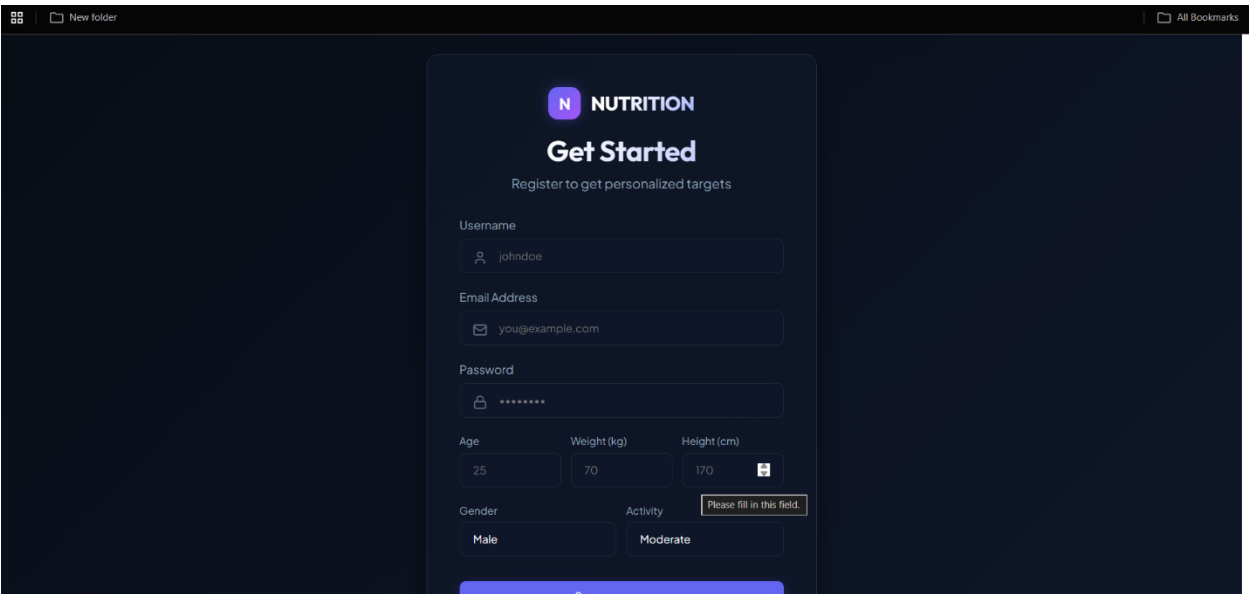
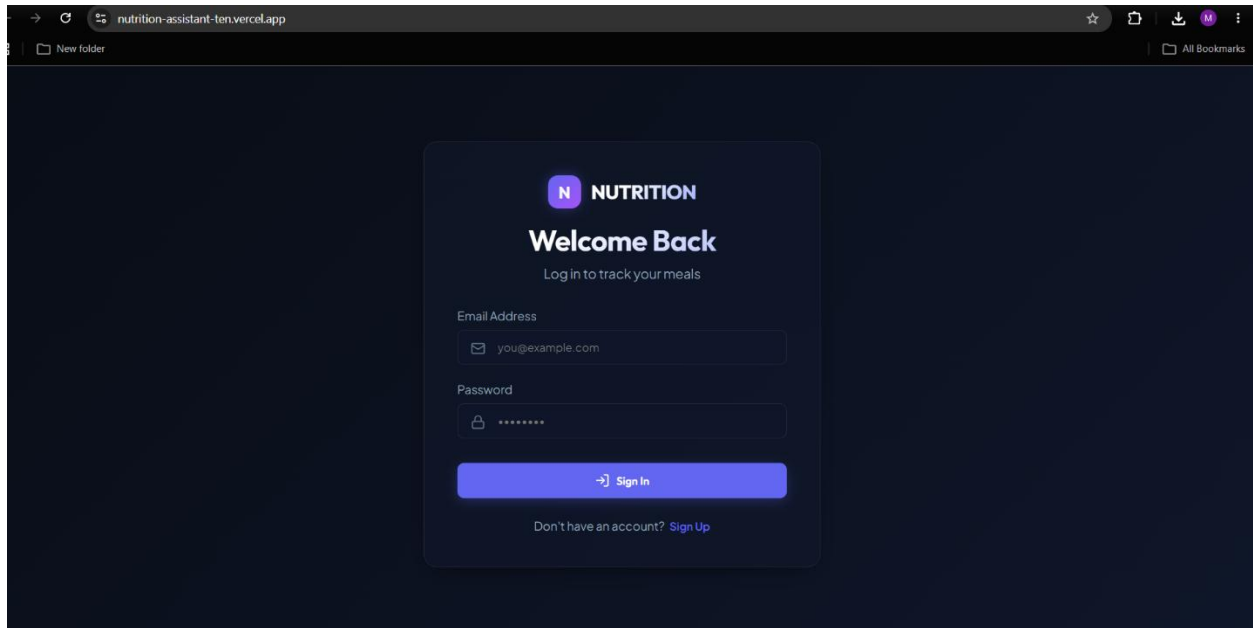


Fig 7.3 – Login Page



8. Advantages & Disadvantages

Advantages

- Gives users a fast, reliable way to search food and understand its nutritional value.
- Automatically calculates calorie and macronutrient totals, removing manual math.
- Interactive dashboard makes daily progress easy to understand at a glance.
- Clean, modular MVC architecture makes the system easy to maintain and extend.
- Cloud-hosted database (MongoDB Atlas) keeps nutrition history safe and accessible.
- Simple, intuitive interface suitable for both nutrition beginners and experienced trackers.

Disadvantages

- Nutrition accuracy depends on the quality and completeness of the underlying food database.
- Does not yet support barcode scanning or photo-based food recognition for faster logging.
- No meal-planning or recommendation engine yet — tracking is currently manual entry only.
- No mobile application yet; the platform is currently accessible only via web browser.
- Does not yet account for individual dietary conditions (e.g., allergies, medical restrictions) in recommendations.

9. Conclusion

Nutrition Assistant successfully delivers a full-stack, easy-to-use nutrition tracking platform built on the MERN stack. By combining secure user authentication, a searchable food database, automatic calorie and macronutrient calculation, and a clear dashboard summary, the application gives users a practical way to build and sustain healthy eating habits. The modular MVC architecture ensured clean separation between the data, business logic, and presentation layers, making the system maintainable, testable, and straightforward to extend. Overall, the project meets its core objective of removing the friction from everyday nutrition tracking.

10. Future Scope

- Add barcode scanning and photo-based food recognition for faster logging.

- Introduce a meal planner with AI-based nutrition recommendations.
- Add weekly and monthly nutrition reports and trend charts.
- Support dietary preferences and restrictions (vegan, allergies, medical conditions) in search and recommendations.
- Develop a mobile application (React Native) for on-the-go tracking.
- Add water intake tracking and reminder notifications.
- Introduce social/community features for sharing progress and healthy recipes.

11. Appendix

Dataset

Nutrition Assistant uses a food/nutrition reference database integrated via the backend Food module rather than a static machine-learning dataset; nutritional values are served through the same data integration referenced in the Food schema and food controller.

Project Demo & Live Links

Live Website: <https://nutrition-assistant-ten.vercel.app/>

Project Demo Video: <https://drive.google.com/file/d/1qSYxZaZ1YlfyhAmkOxF3N0Kg1IWGlVK6/view>

GitHub Repository: <https://github.com/IslavathuSurya-89/Nutrition-Assistant>